Synthetic Control

PS200C, Spring 2026

Chad Hazlett

UCLA

chazlett@ucla.edu

Roadmap

1. Motivation: where DID fails
2. A DGP we can reason about: the linear factor model
3. Intuition: balance on Y^{pre} buys balance on $Y(0)_{\text{post}}$
4. Prop 99: a first pass
5. What can go wrong: EIV & recoverability
6. Flavors of synth and a bias-decomposition view
7. Practice and inference

Where DID gets uncomfortable

DID (and staggered DID) require no time-varying confounding, conditional on unit and time FEs.

That is what “parallel trends” encodes.

E.g. In 1989 California’s Prop 99 went into effect, raising cigarette taxes. You want to see if it reduced cigarette sales.

What changes over time, is felt differently in different states, and is related to the outcome and treatment chances? E.g.

- Health consciousness
- Anti-tobacco sentiment
- Willingness to use tax structure to achieve social aims

The hope

Synthetic control offers a way to address time-varying unobserved confounding by leaning on the shape of pre-treatment outcomes to detect the confounding signal.

Intuition: If you had a state with same pre-treatment trajectory of cigarette sales (Y_i^{pre}), then it must have similar values on (time-varying) confounders. Further we can just "upweight" states with similar Y_i^{pre}.

This is seductive and popular: Athey & Imbens (2017) call it "arguably the most important innovation in the policy evaluation literature in the last 15 years," and the original Abadie–Diamond–Hainmueller (2010) paper has 9k citations.

... but I've never seen identification assumptions discussed in an application. When does it actually work? What are the required assumptions?

Heads-up: synth is <u>not</u> DID

This should be obvious now but in case you missed it...

- Synth does **not** assume parallel trends.
- A common student error is to think of synth as "a better DID." It isn't. It's a different identification strategy.
- In fact, once a synth weighting matches the treated unit at the last pre-treatment period, <u>DID would just become the diff.</u>

Demystifying: it's mean-balancing

Most synthetic-control estimators (Abadie's OG, augmented SCM, synthetic DID, `tjbal` with linear kernel) effectively: **find donor weights w that make the weighted control's Y^{pre} match the treated unit's Y^{pre}.**

This is the same cross-sectional weighting/matching you already know — on the wide-form panel where "X" is the vector of pre-treatment outcomes $(Y_{i,1}, \ldots, Y_{i,T_0})$.

What changes between flavors:

- what we balance (means? richer features?),
- what constraints on w (simplex? entropy? ridge?),
- whether we also weight time periods (SDID).

Pretend we believe the linear factor model

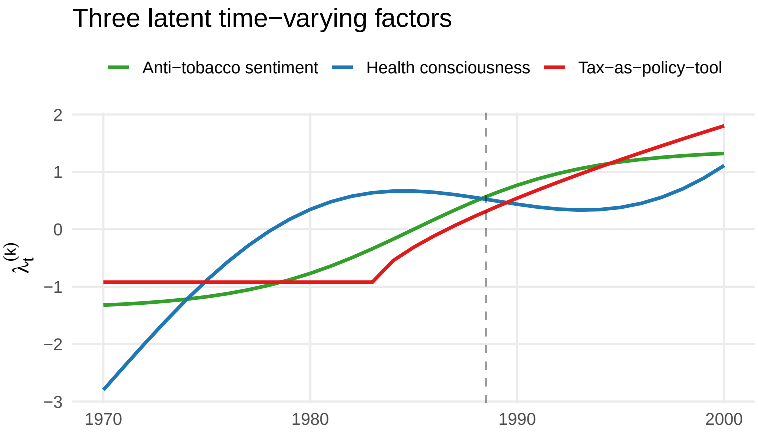
$$Y_{it}(0) \;=\; \alpha_i + \lambda_t^\top u_i + \varepsilon_{it}$$

- $u_i \in \mathbb{R}^K$: unit-level **factor loadings** (time-invariant)
- $\lambda_t \in \mathbb{R}^K$: **time-varying factors** (common across units)
- ε_{it}: idiosyncratic noise

This is a vivid DGP you can imagine two ways:

- Unit confounding characteristics (u_i) whose impact vary over time (λ_t).
- Or time-varying “signals” whose influence on unit *i* depends on “loadings” u_i.

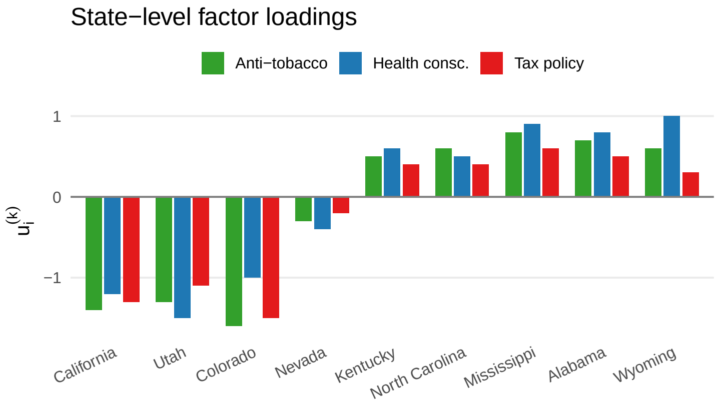
Three latent factors $\lambda_t^{(k)}$ for CA Prop 99



Smooth time-varying signals: health consciousness, anti-tobacco sentiment, tax-as-policy-tool.

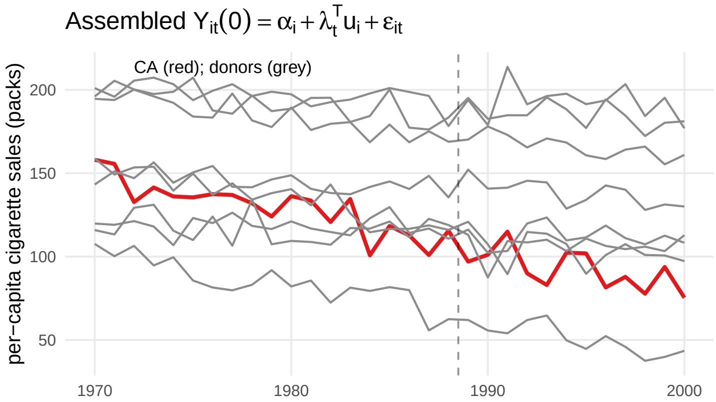
State-level loadings $u_i^{(k)}$

Imagine state-level loadings (u_i) on these such as,



CA, UT, CO, NV have high loadings on all three.
KY/NC/MS/AL/WY quite different.

Assembled $Y_{it}(0) = \lambda_t^\top u_i + \varepsilon_{it}$



Different states evolve differently — not because λ_t differs, but because their loadings u_i project the same factors differently.

The estimand and the obstacle

Estimand: the sample ATT for CA,

$$\tau_{\text{ATT}} \;=\; \mathbb{E}\big[\, Y_{i,T+1}(1) - Y_{i,T+1}(0) \;\big|\; i = \text{CA} \,\big]\,.$$

Why DID/FE won't do it:

- u_i is time-invariant — TWFE can wipe its mean effect
- but the differential exposure λ_t is time-varying
- TWFE leaves $(\lambda_t - \bar{\lambda})^\top u_i$ behind — exactly the confounder

We need to neutralize u_i itself, or at least its post-treatment imprint $\lambda_{T+1}^\top u_i$.

Connection back to the DID lecture

We saw at the end of DID:

> *"Ensuring balance on prior outcomes starts to bring you toward other strategies that effectively condition on Y^{pre}, e.g. synthetic control."*

That's the bridge we're crossing today.

Intuition: Y^{pre} encodes u_i

The pre-treatment trajectory of unit i:

$$Y_{i,\text{pre}} \;=\; \Lambda_{\text{pre}}\, u_i + \varepsilon_{i,\text{pre}}, \qquad \Lambda_{\text{pre}} = \begin{bmatrix} \lambda_1^\top \\ \vdots \\ \lambda_T^\top \end{bmatrix}$$

If Λ_{pre} is rich enough, similar $Y^{\text{pre}} \Rightarrow$ similar u_i.

And similar $u_i \Rightarrow$ similar $\lambda_{T+1}^\top u_i =$ similar $Y(0)_{T+1}$.

That's our identification chain.

The chain

So we're hoping:

balance on $Y_i^{\text{pre}} \to$ balance on $u_i \to$ balance on $Y(0) \to$ sATT

Plan: (1) Find donor weights w such that

$$\sum_{i \in \text{donors}} w_i \, Y_{i,\text{pre}} \;\approx\; Y_{\text{CA},\text{pre}}.$$

(2) Get weights by choosing, with some constraints/penalties:

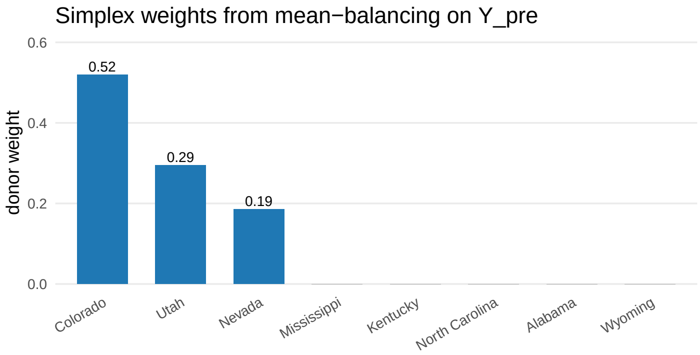
$$\widehat{Y(0)}_{\text{CA},T+1} = \sum_i w_i \, Y_{i,T+1}$$

(3) Then estimate sATT with:

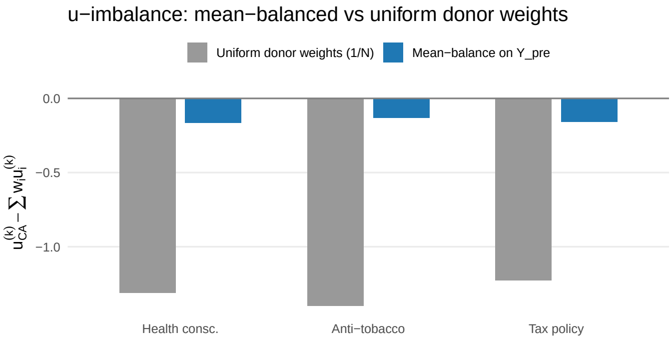
$$\widehat{\tau} = Y_{\text{CA},T+1} - \widehat{Y(0)}_{\text{CA},T+1}$$

Sanity check: which donors get picked?

Toy LFM, 8 donors (UT, CO, NV + KY, NC, MS, AL, WY). Solve for simplex weights **w** matching CA's Y_i^{pre}:



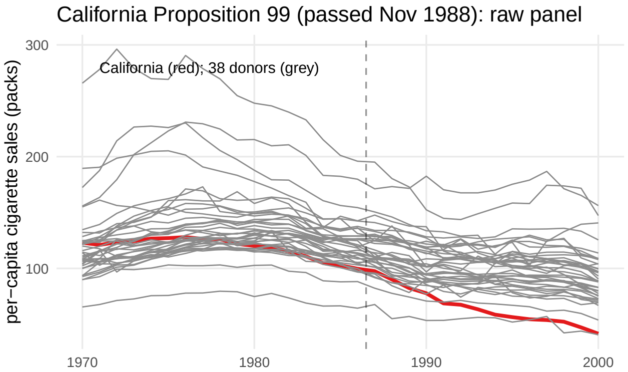
Does it balance on confounding characteristics (u)?



Implied bias on $Y(0)_{T+1}$:

- Uniform weights: bias $= \text{imbal}(u_i)^\top \lambda_{T+1} \approx$ **19** packs/capita
- Mean-balance: bias $= \text{imbal}(u_i)^\top \lambda_{T+1} \approx$ **0.7** packs/capita

California Proposition 99, real data now



- Per-capita cig. sales, 1970–2000, CA (red) vs. 38 donor states (grey).
- I use **1987 as the first treated period**. ADH used 1989.[1]

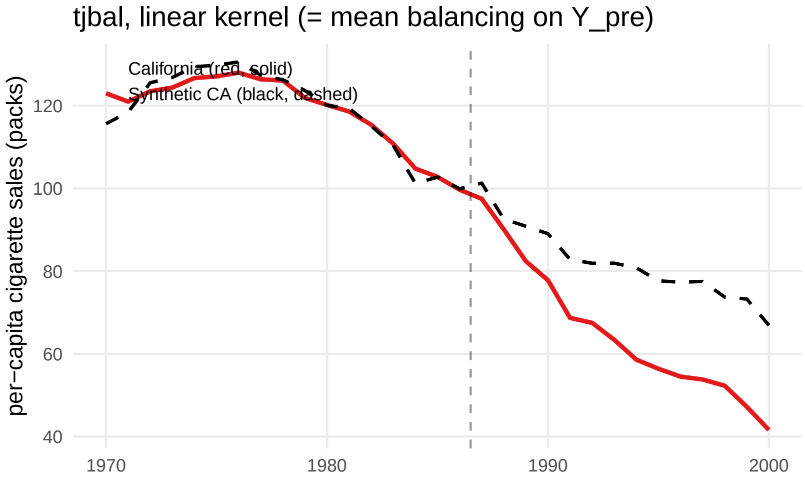
[1] 1987: Proposal made; coalition formed, initiative drafted; 1.1M signatures gathered late-1987 through May 1988; ballot passed Nov 1988; tax effective Jan 1989. So: anticipatory purchasing plausible in 1987-1988.

The tool: `tjbal`, linear kernel = mean balancing

```
library(tjbal)
fit_lin <- tjbal(
  data    = df,
  Y       = "cigsale",
  D       = "D",
  index   = c("state", "year"),
  demean = FALSE,
  kernel = FALSE   # FALSE = linear = mean balance on Y_pre
)
```

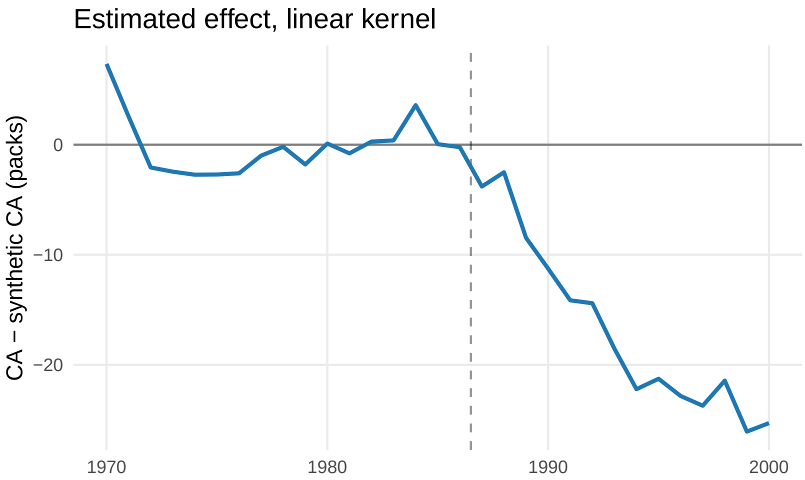
That's the whole call. We'll flip `kernel = TRUE` later.

Synthetic California (linear kernel)



Pre-period fit looks good. Post-period gap: CA falls below the synthetic.

Estimated effect (linear kernel)



Average post-tx ATT ≈ -17 packs per capita per year (1987 onward).
(Original SCM gives ≈ -19)

Worry 1: EIV bias

Two main problems here. The first is **errors in variables** (EIV): Balancing on a noisy version of u_i leaves residual imbalance in the true u_i.

Bias: scales with σ_ε relative to signal.

Shrinks with T: more pre-periods = better-resolved u_i (per-state averages of ε shrink as $1/\sqrt{T}$).

This is a finite-sample problem. Data fixes it eventually.

The next problem is not so kind.

Worry 2: Recoverability

Pretend there is no noise, $Y_i^{\text{pre}} = \Lambda_{\text{pre}} u_i$

When does balance on $\Lambda_{\text{pre}} u_i$ buy you balance on u_i?

Formally: λ_{T+1} must lie in the row span of Λ_{pre} — a **rank condition** required on top of the LFM.

In plain English: every confounder that affects post-treatment outcomes must leave a footprint in Y_i^{pre} that's distinguishable from other confounders' footprints.[2]

This is an **identification-level** assumption, not a simple regularity condition. More N helps estimation, but doesn't fix it.

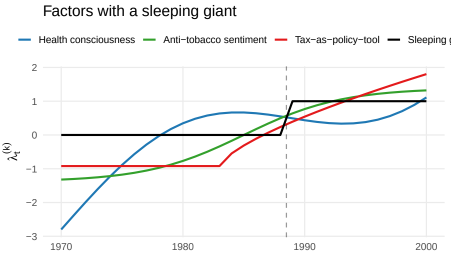
[2] The rank condition, LPO (post-outcome is a linear function of Y_i^{pre} with the same coefficients across units), and the "distinguishable footprints" idea are three views of the same condition, with the same failure modes. See Appendix for a trip down the rabbit hole.

Concretely, what breaks recoverability?

As usual, the counterexamples help us see clearly.

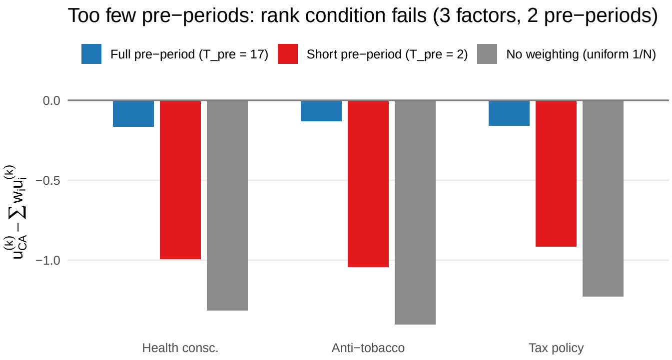
Counter 1: Pre-period too short to disentangle the confounders. With K confounders but only $T_{\text{pre}} < K$ pre-periods, distinct confounders leave indistinguishable footprints — Y_i^{pre} literally can't tell them apart.

Counter 2: Imbalanced sleeping giants. A confounder with little variation before treatment, more after, AND whose u_i is imbalanced across treatment arms. The pre-period leaves no footprint for it at all — it's silent. Then it wakes up at $T+1$. I.e. something newly changing in 1987 onward, felt more by CA.



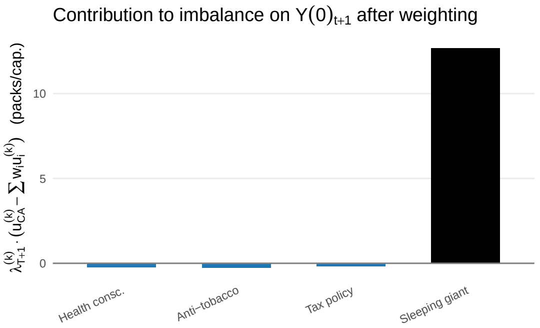
Insufficient periods demo: 3 factors, only 2 pre-periods

Keep the same 3-factor LFM, but balance only on Y_i^{pre} from 1985 and 1986 (so $T_{\text{pre}} = 2 < K = 3$). The simplex weights perfectly match those two years — but the rank condition fails, so balance on u_i does not follow:



Sleeping giant demo: Uh oh!

- Pre-treatment period is uninformative about $u^{(4)}$
- Suppose the 4th factor has $u^{(4)}_{CA} = 1$ while donors have $u_i \sim \mathrm{Unif}(-0.5, 0.5)$.



Any factor that's invisible in the pre-period leaves no trace to balance on...effectively unobserved confounding.

Aside: lagged-DV regression does the same thing

LDV regression:

$$Y_{i,T+1} \;=\; \alpha + \beta^\top Y_i^{\text{pre}} + \tau D_i + \varepsilon_i.$$

By FWL, $\hat{\tau} = \mathrm{Cov}(Y_{i,T+1}^{\perp Y_i^{\text{pre}}}, D_i^{\perp Y_i^{\text{pre}}})/\mathrm{Var}(D_i^{\perp Y_i^{\text{pre}}})$.

...i.e. anything linear in Y_i^{pre} is removed, just as mean balancing on Y_i^{pre} removes anything linear in it from ATT.

Where they differ: the nature of the implied weights.

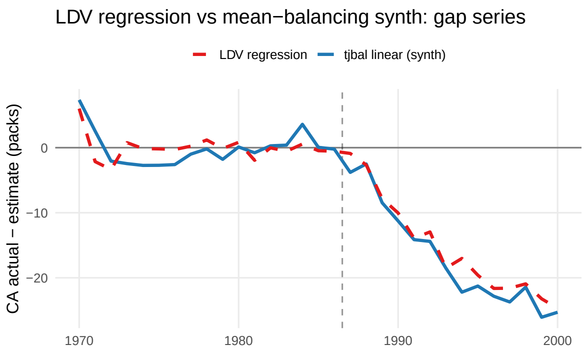
- synth: sparse, $w_i \geq 0$, $\sum w_i = 1$ — may fail to find balance to varying degrees, but refuses extrapolation.
- LDV (OLS): never visibly fails, but the equivalent weights can be negative, indicating extrapolation gambles.[3]

[3] Crazy but true: with exactly one treated unit, the LDV gives you same answer as imputing/g-comp that trains on controls only, predicts on treated and contrasts with observed treated outcome. Does not hold with more treated units.

Back to Prop 99: LDV ≈ mean-balancing synth

Fit this and plot $\hat{\tau}_{t^*}$ in each post-period,[4]

$$Y_{i,t^*} \;=\; \alpha + \beta^\top Y_{i,1970:1986} + \tau_{t^*} D_i + \varepsilon_i,$$



[4] For pre-period t^*, drop t^* from the regressors (leave-one-out placebo).

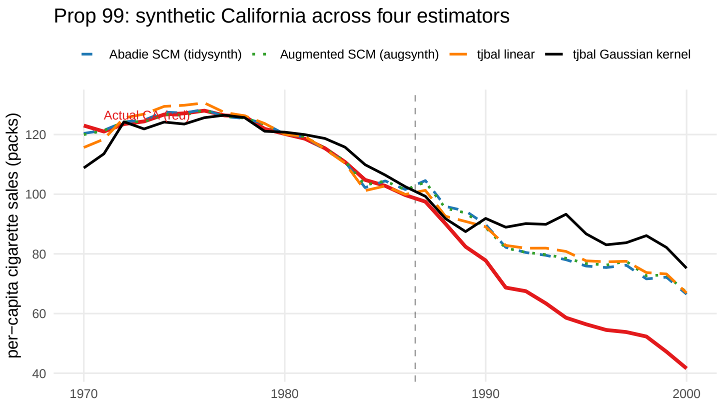
The family of synth estimators

All are weighted-control estimators differing in what they balance:

- **Classical SCM (Abadie):** mean-balance Y^{pre} (and optional covariates) on the simplex
- **Augmented SCM (Ben-Michael et al.):** SCM + outcome-model bias correction
- **Synthetic DID (Arkhangelsky et al.):** unit weights to balance "who," time-weights to anchor a DID-on-synth step; ridge regularization
- **tjbal (Hazlett, Xu):** balance kernel features of Y^{pre}; captures slope/curvature, scale-invariant
- ... and many others.

While some of these add in fallback, the core identification concerns are relevant to all.

Prop 99 across four estimators



Post-treatment average ATT (1987 onward):

- Abadie SCM (tidysynth): ≈ -16.5
- Augmented SCM (augsynth): ≈ -16.7
- tjbal linear: ≈ -16.9
- tjbal Gaussian kernel: ≈ -22.7

Broader view: Bias decomposition under general setting

Instead of imposing a DGP, doing mean balancing, and asking if it "is identified," consider the weakest setting and broader class of balancing estimators and just look at bias terms.[5]

- Assume only latent ignorability: $Y_i(0) \perp\!\!\!\perp D_i \mid u_i$
- And weakest compatible DGP: $Y_i(0) = g(u_i) + \varepsilon_i$.
- And that you balance on *some* (possibly nonlinear) features $\phi(Y_i^{\text{pre}})$.

[5] Following the `tjbal` manuscript (Asher, Hazlett, Xu, in progress); active area of research.

Broader view: Bias decomposition under general setting

Under this, the bias of the synth-type estimator decomposes as:

$$\widehat{\tau} - \tau \approx \underbrace{r_{\text{bal}}}_{\text{balancing}} + \underbrace{r_{\text{eiv}}}_{\text{EIV}} + \underbrace{r_{\text{rec}}}_{\text{recoverability}} + \underbrace{\varepsilon_{T+1}}_{\text{post-noise}}$$

- r_{bal} How well your weights actually achieve balance on the chosen features. Shrinks with N and a good optimizer.
- r_{eiv} Y_i^{pre} is a noisy view of u_i. Shrinks as $T_{\text{pre}} \to \infty$, but worry-worthy in finite T.
- r_{rec} Confounding not captured by $\phi(Y_i^{\text{pre}})$, i.e. bias due to sleeping giants **does not shrink with data**.
- ε_{T+1} Mean-zero; doesn't bias but adds variance (scales with $\|\hat{w}\|^2$).

Tradeoffs: choice of ϕ may help ensure r_{rec} isn't due to specification in short T, but can worsen r_{bal} and r_{eiv}.

Practical considerations:

- Taking care with the pre-treatment window: You want more T, but emphasize time when the relevant confounders are awake (so, usually more recent).
- **Treatment-year coding.** Watch out for anticipation, as relevant in the Prop 99 case.
- **Multiple treated units, staggered adoption.** Fine, with care to use correct donor pool at each wave. (e.g. `tjbal` handles automatically)
- **Pre-fit quality**: how well does $\sum w_i Y_{i,\text{pre}}$ track $Y_{\text{CA,pre}}$? Failure here is failure to even condition on Y_i^{pre}.
- **Placebo-in-time**: shrink the pre-window, "predict" the held-out periods. Detects recoverability problems within the pre-period.
- **Placebo-in-space**: re-estimate with each donor as "treated." Tells you the spread of placebo effects. Can especially check on "units most likely to have been treated" per some propensity score model.
- **Weight inspection**: For sanity check, interpretation, and because extreme weights (high concentration) are brittle.

Inference (briefly)

- **One treated unit**: genuinely hard. No nice CLT. Permutation/placebo plots are the default. But could simply use LDV.
- ≥ 2 **treated units**: jackknife works (the tjbal default).

The honest summary: in synth, the point estimate is usually the work, and inference is a sanity check rather than a guarantee.

Wrap-up

Synthetic control is not a panacea, of course.

- It does avoid parallel trends / no time-varying confounders. YAY!
- Instead hopes for *recoverability*: Y_i^{pre} encodes sufficient evidence for the "types" of units (u_i) so that conditioning on Y_i^{pre} is as good as conditioning on u_i.
  - Enough periods, no (imbalanced) sleeping giants.
- On top of this, estimation grapples with:
  - EIV bias
  - What notion of similarity in Y_i^{pre} encodes similarity in u.

Wrap-up continued

Where does that leave you?

- More periods is good. Imbalance on Y_i^{pre} after weighting is pretty bad.
- Try mean balance first, then consider higher order if you can afford it.
- Be humble, as usual...as ever, the procedure doesn't make the result causal.

Some reasons for hope (speculative):

1. If you get decent mean balance on Y_i^{pre} then you are getting DID for free (or use sDID etc.), but you are at least adding balance on whatever is linear in $\phi(Y_i^{\text{pre}})$.
2. You might be able to argue that time-varying confounders are *smooth* and so cannot be sharp sleeping giants.
   - Use domain expertise: why the treated units took treatment when they did, and what realistically influences Y_t at scale...can help rule out the "suddenness" of a sleeping giant.
3. Placebos tests can help ensure you are at least catching the structure that exists prior to treatment, leaving you with genuine sleeping giants to worry about.

Appendix: rank condition = LPO

We have three views to tie together:

- Formal: λ_{T+1} lies in the row span of Λ_{pre}.
- LPO: $Y_{i,T+1}(0)$ is a linear function of Y_i^{pre} — same coefficients for every unit.
- Footprints: every confounder leaves a distinguishable trace in Y_i^{pre}.

Absent noise, these are formally equivalent...

Reminder: LPO (Linearity in Prior Outcomes)

LPO is a weaker hope than full LFM:

$$Y_{i,T+1}(0) \;=\; \alpha + \beta^\top Y_i^{\text{pre}} + \eta_i, \qquad \mathbb{E}[\eta_i \mid D_i] = 0.$$

Just: each unit's noiseless post outcome is a linear function of its pre-period trajectory, with the same (α, β) across units.

Under LPO, weights $\mathbf{w}$ with $\sum_i w_i Y_i^{\text{pre}} = Y_{\text{CA}}^{\text{pre}}$ give

$$\sum_i w_i\, Y_{i,T+1}(0) \;=\; \alpha + \beta^\top Y_{\text{CA}}^{\text{pre}} + \sum_i w_i\, \eta_i \;\approx\; Y_{\text{CA},T+1}(0).$$

So LPO is what makes mean-balancing on Y_i^{pre} buy you balance on $Y(0)_{T+1}$.

Rank condition ⇔ LPO (under noiseless LFM)

Rank condition: there exist coefficients $c_1, \ldots, c_{T_{\text{pre}}}$ such that

$$\lambda_{T+1} = \sum_{t=1}^{T_{\text{pre}}} c_t \, \lambda_t .$$

Multiply both sides by $u_i^\top$:

$$\underbrace{\lambda_{T+1}^\top u_i}_{= Y_{i,T+1}(0)} = \sum_t c_t \underbrace{\lambda_t^\top u_i}_{= Y_{it}(0)} = c^\top Y_i^{\text{pre}} .$$

Every unit's post outcome is the same linear function of its Y_i^{pre} — exactly LPO.

Two views, one condition:

- Factor side: post-period λ extrapolates linearly from pre-period λ's.
- Outcome side: post-period $Y(0)$ extrapolates linearly from pre-period Y.

And the “distinguishable footprints” picture

A confounder’s **footprint** in Y_i^{pre} is its column of Λ_{pre} — the pattern of factor weights $\lambda_t^{(k)}$ over the pre-period.

If every confounder leaves a linearly distinguishable footprint, the rows λ_t span all K dimensions of $\mathbb{R}^K$, so the row span contains every possible λ_{T+1}. Rank condition holds. LPO holds.

Two failure modes through this lens:

- **Sleeping giant**: one column of Λ_{pre} is all zeros (factor silent pre-treatment). Row span has no component along that coordinate; any λ_{T+1} that loads on it sits outside. LPO fails.
- **Too few pre-periods** ($T_{\text{pre}} < K$): row span has dimension at most $T_{\text{pre}} < K$; can’t cover all of $\mathbb{R}^K$. Most λ_{T+1} sit partially outside. LPO fails.